When Bad News Become Good News

Towards Usable Instances of Learning with Physical Errors

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Learning problems



Learning problems have proven to be interesting computationally hard problems.

- **LPN**
- **LWE**
- **LWR**
- **MLWE**

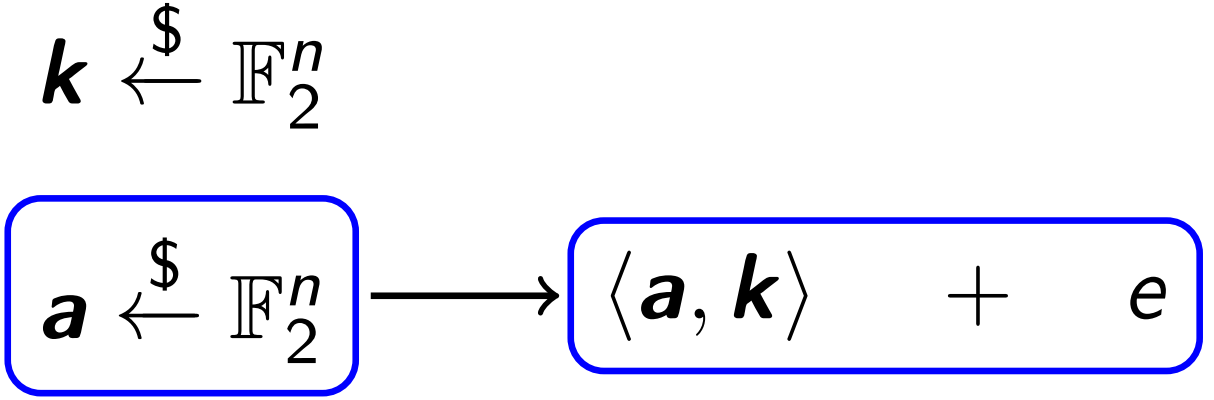
...

$\Rightarrow$

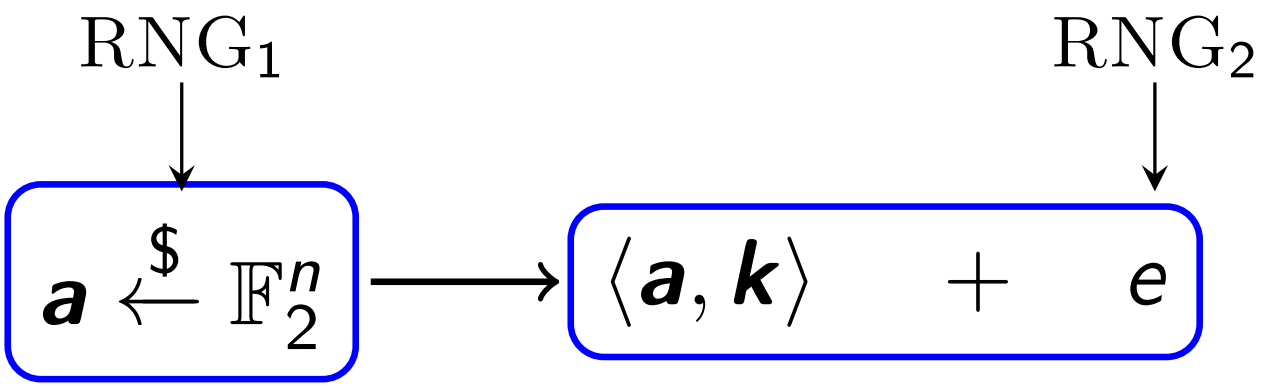
- *One-way function*
- *Secret-key encryption scheme*
- *Post-quantum PKE*
- *Identity-based encryption*
- *Secure MPC*
- *Indistinguishability obfuscation*

...

Learning problems - Presentation

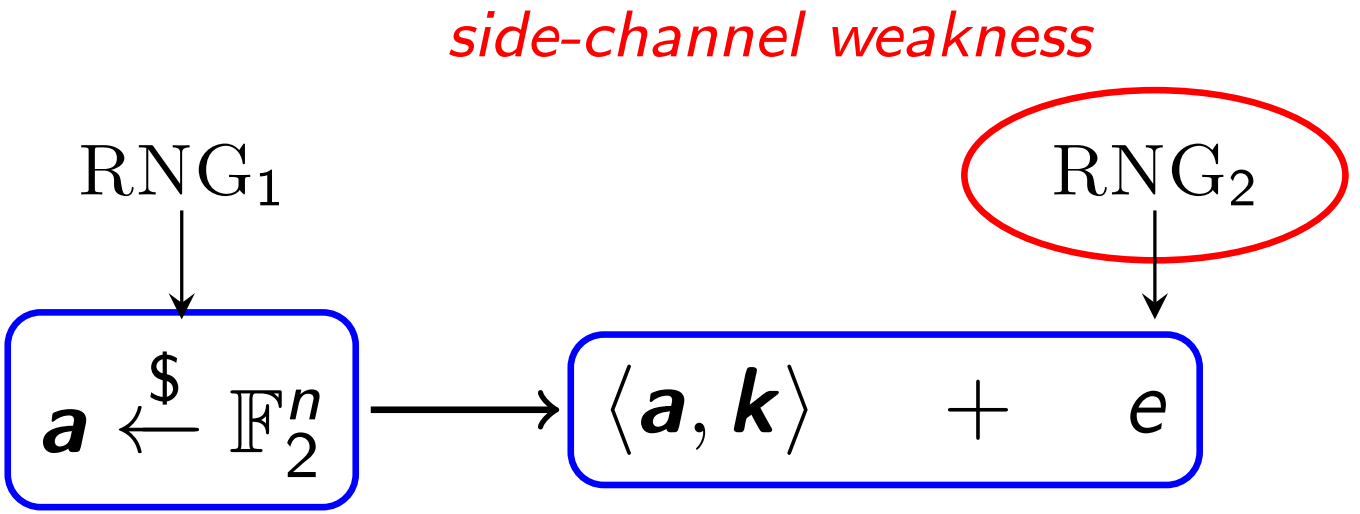


Learning problems - Implementation



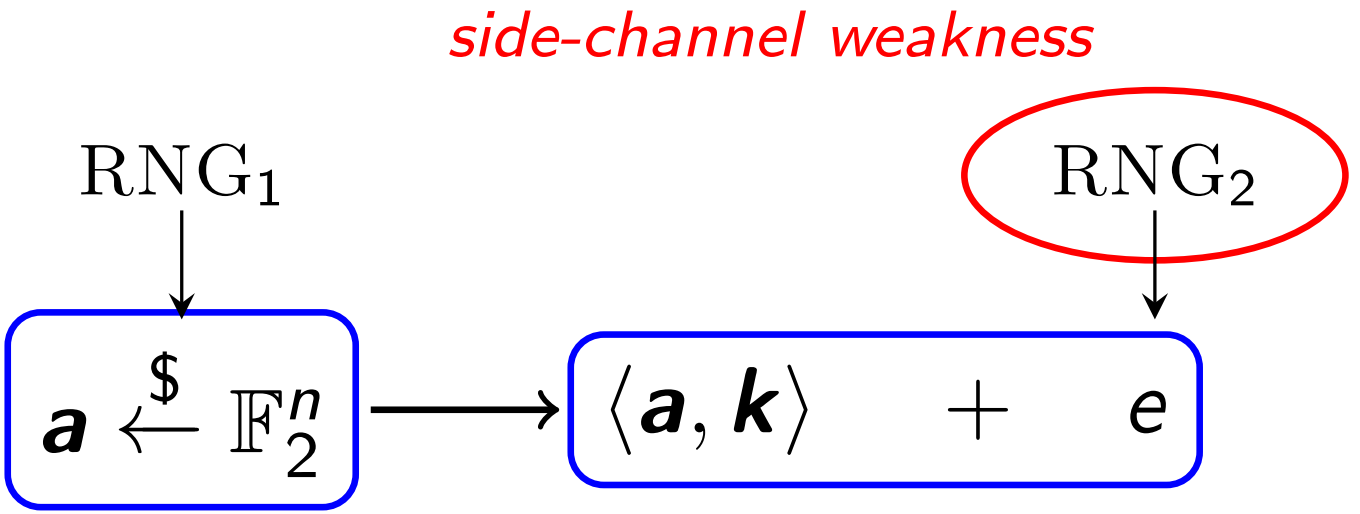
LPN samples

Learning problems - Implementation



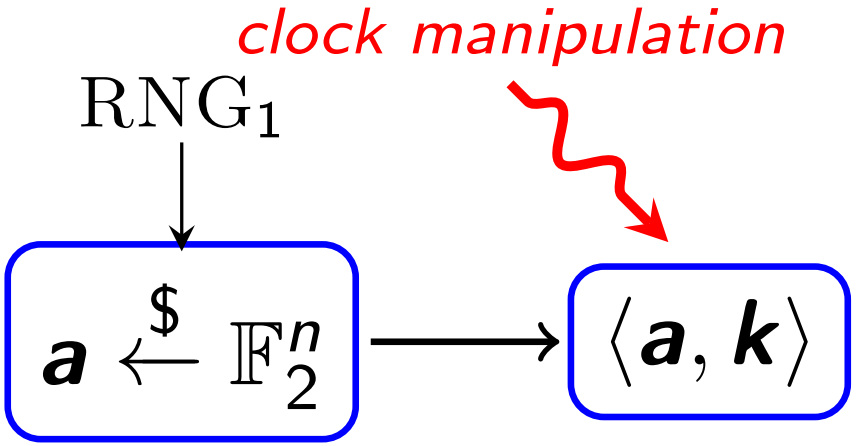
LPN samples

Physical learning problems - Presentation



LPN samples

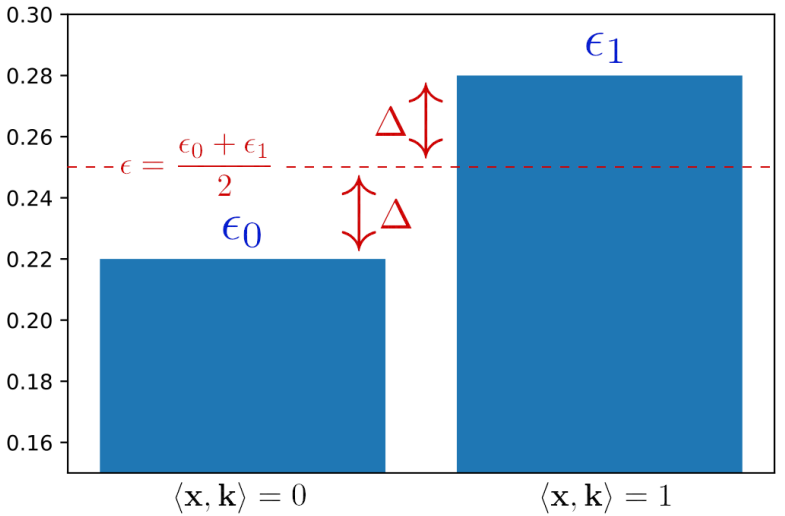
Physical learning problems - Presentation



LPN samples

Physical learning problems - Data dependencies

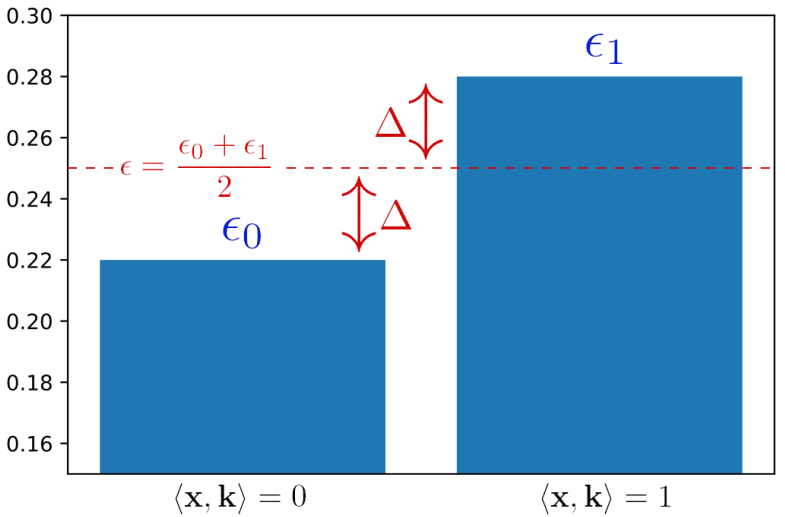
Output data dependencies:



Error probability depending on the correct output value

Physical learning problems - Data dependencies

Output data dependencies:

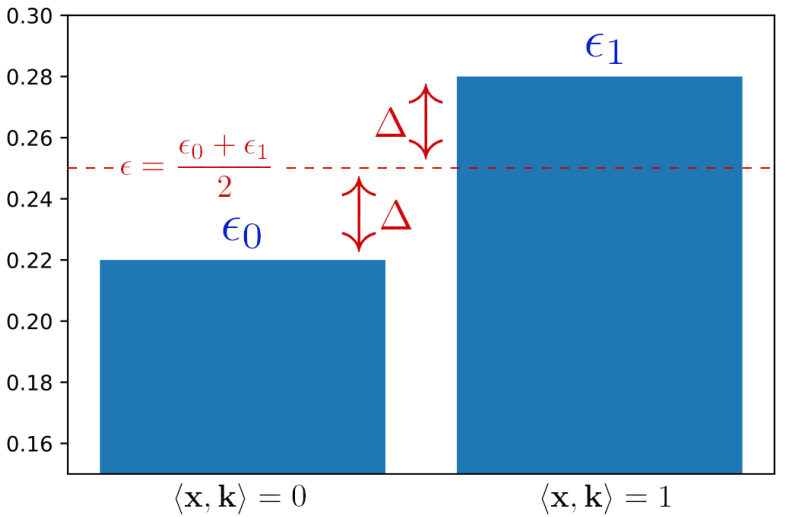


Error probability depending on the correct output value

- Not negligible
- Reduction for LPPN

Physical learning problems - Data dependencies

Output data dependencies:



Error probability depending on the correct output value

- Not negligible
- Reduction for LPPN

Input data dependencies:

- Computationally hard to exploit
- Can be made small by design

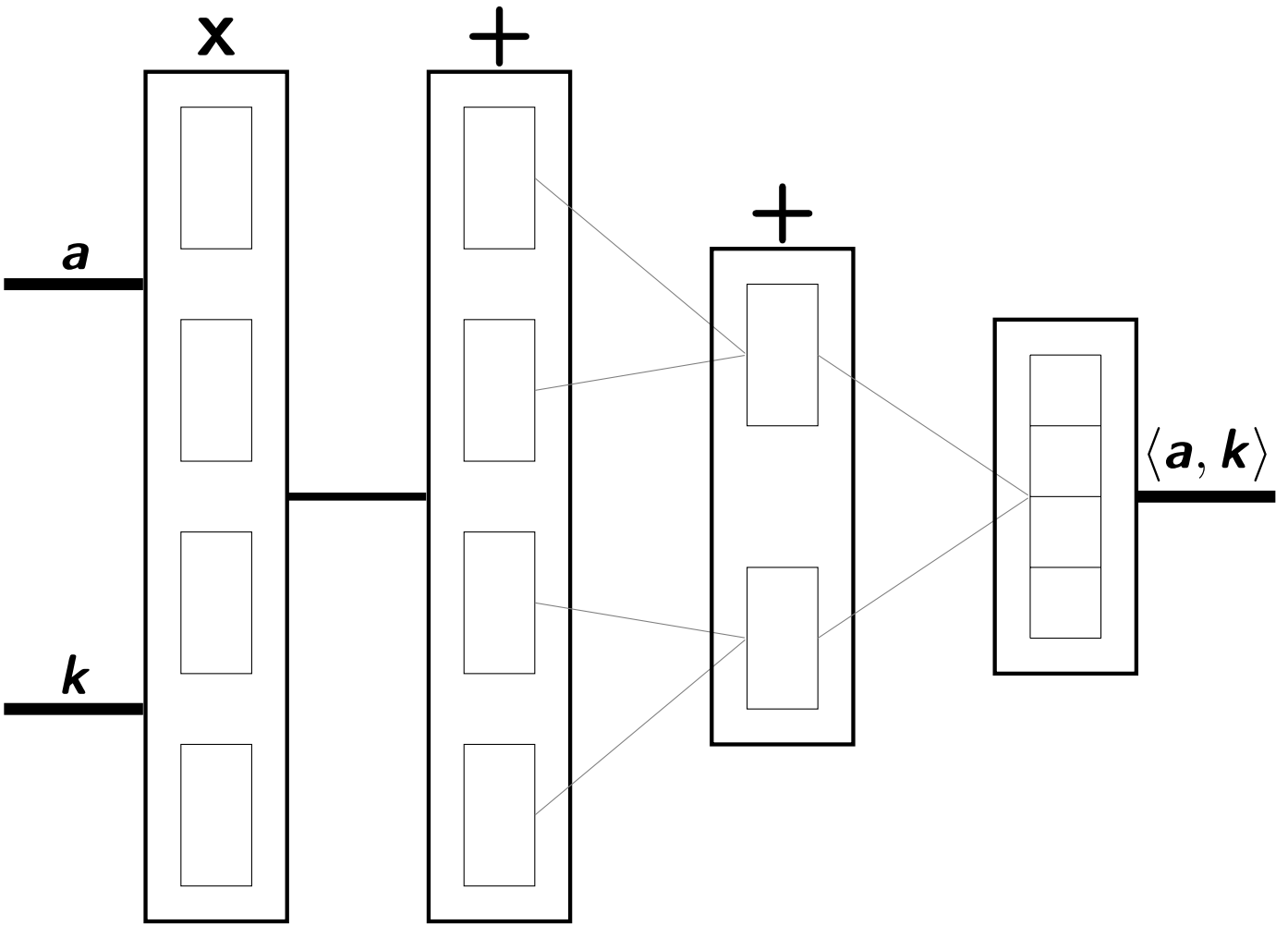
Can this extend to LWE ?



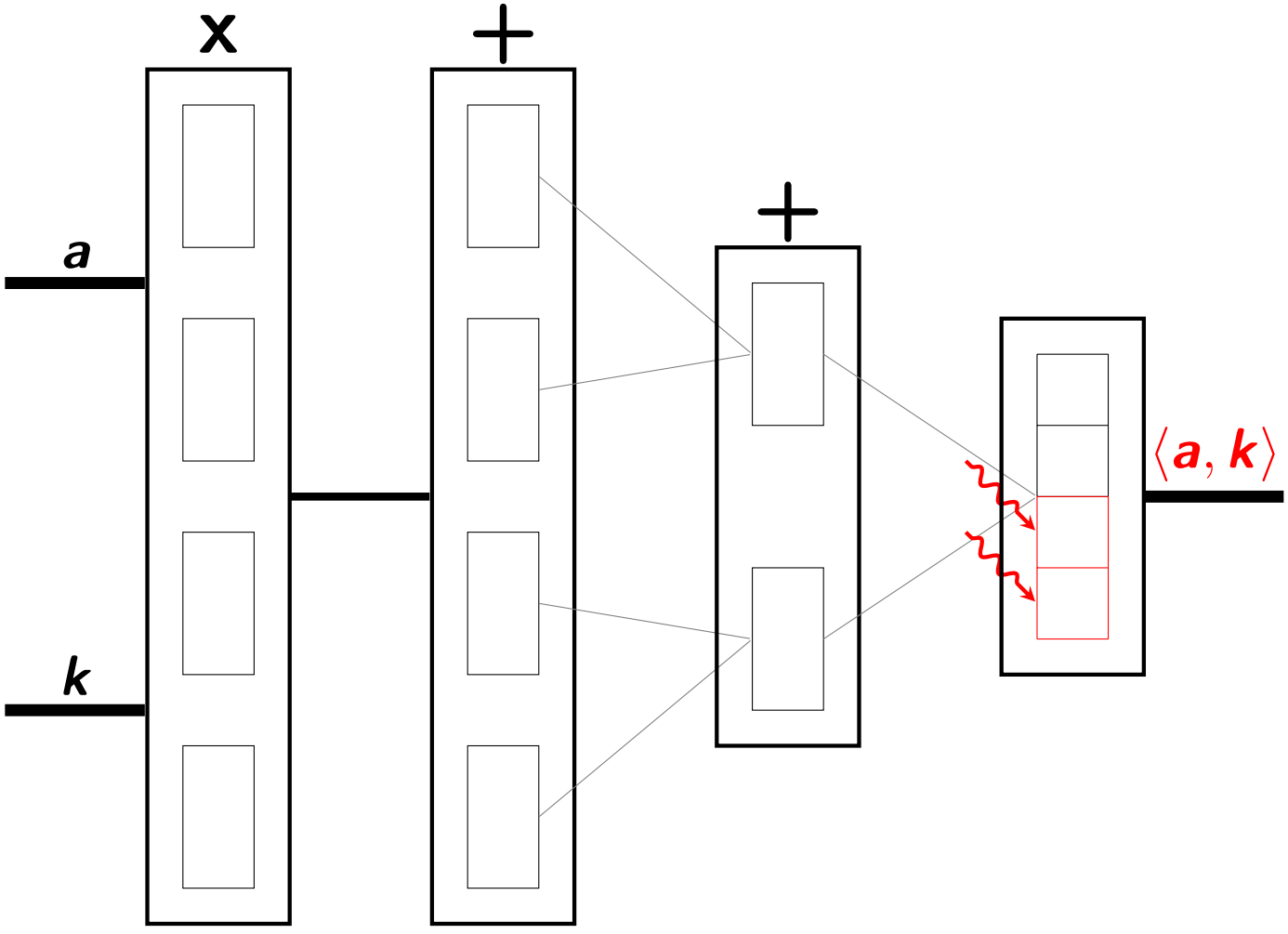
GOAL:

- From **1-bit** to **128-byte** elements physical inner product
- Error distribution approximating a $\mathrm{CBD}_{2 \text{ or } 3}$ (used in Kyber)

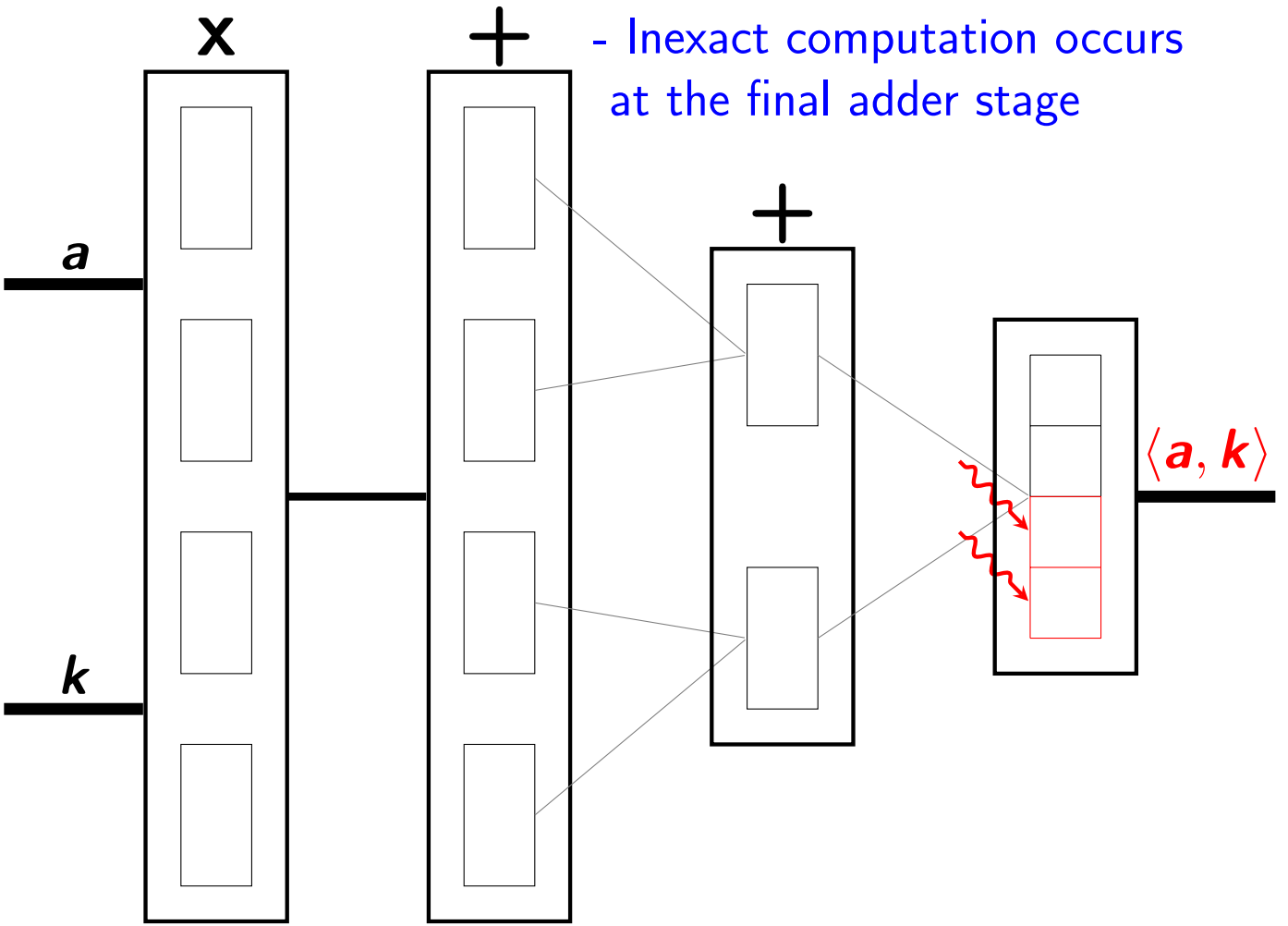
Inner product structure



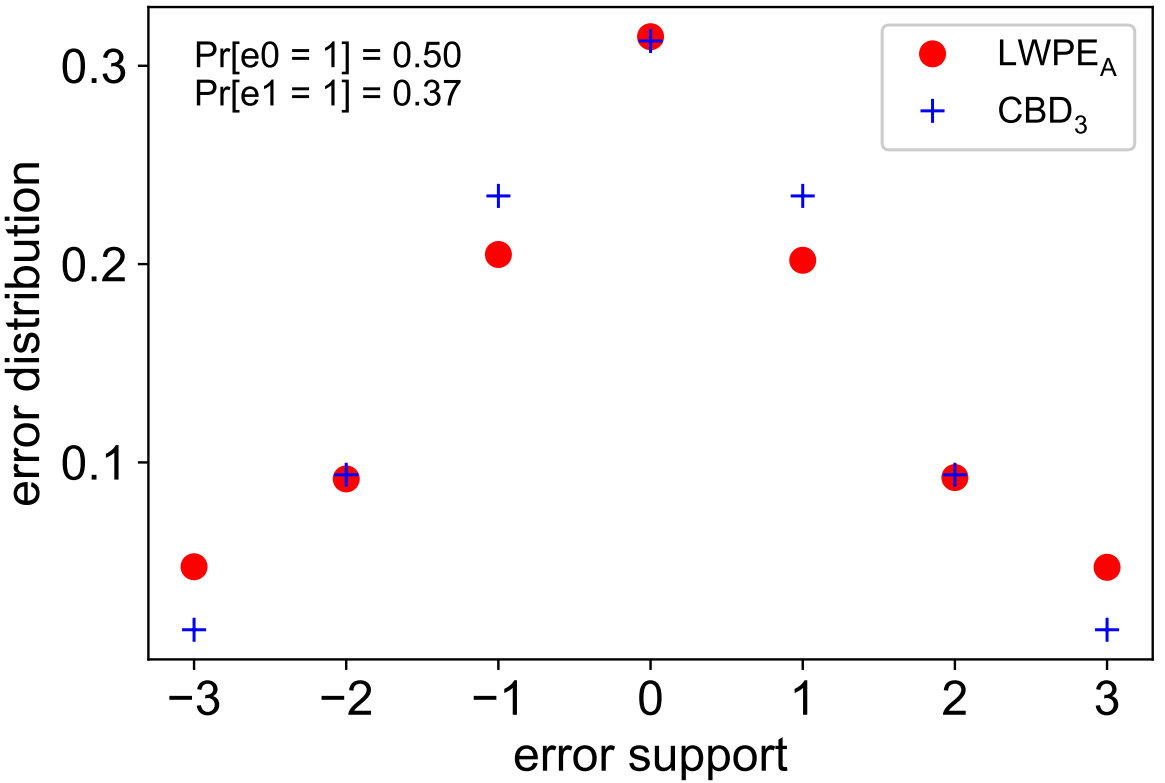
Natural solution - Presentation



Natural solution - Presentation



Natural solution - Results



LWPE$_A$: Error distribution approximating CBD_3.

Bad news - Mathematical data dependencies (1/2)



TOY EXAMPLE:

- $\langle \boldsymbol{a}, \boldsymbol{k} \rangle = 0$
- modulo 4

Bad news - Mathematical data dependencies (1/2)



TOY EXAMPLE:

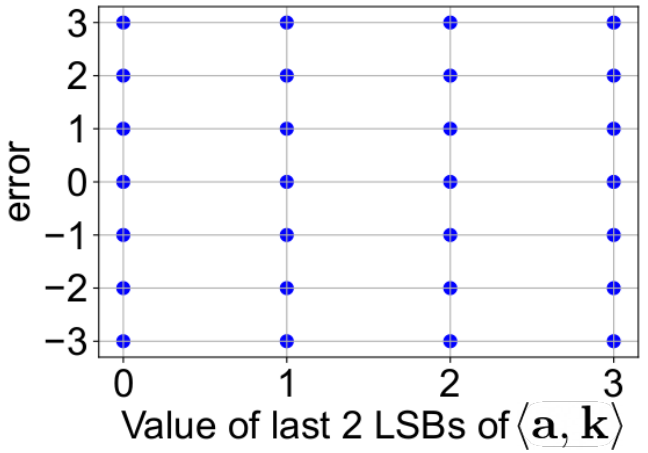
- $\langle \boldsymbol{a}, \boldsymbol{k} \rangle = 0$
- modulo 4

wrong support

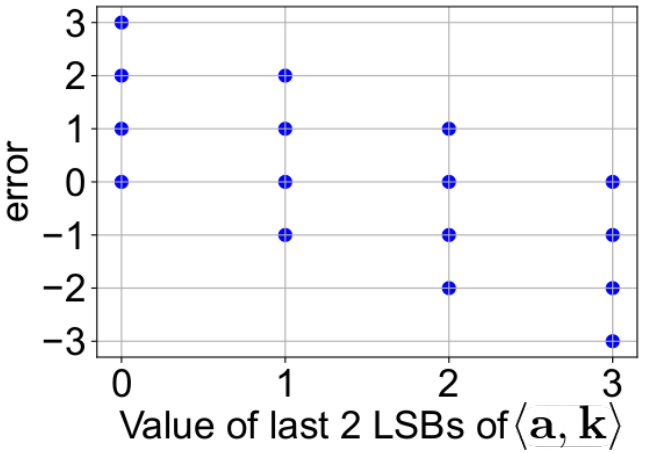
$$\langle \boldsymbol{a}, \boldsymbol{k} \rangle = \overline{00} \longrightarrow \begin{cases} \overline{00} \rightarrow e = 0 \\ \overline{01} \rightarrow e = 1 \\ \overline{10} \rightarrow e = 2 \\ \overline{11} \rightarrow e = 3 \end{cases}$$

Bad news - Mathematical data dependencies

Regular LWE



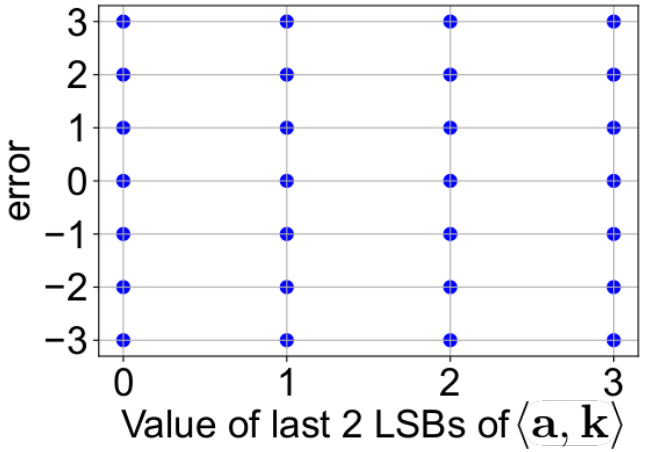
LWPE$_A$

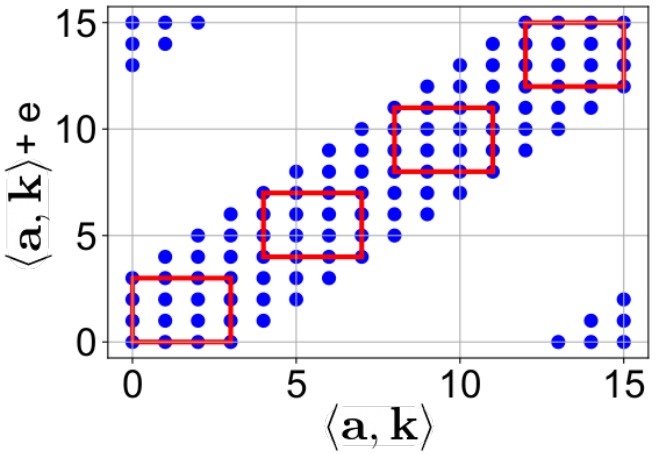


Bad news - Mathematical data dependencies

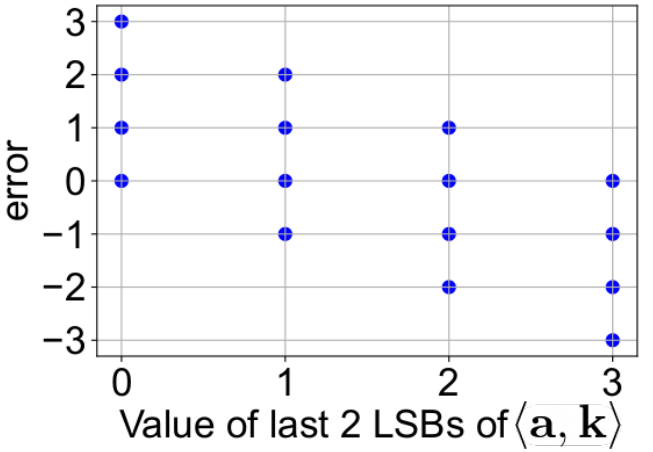


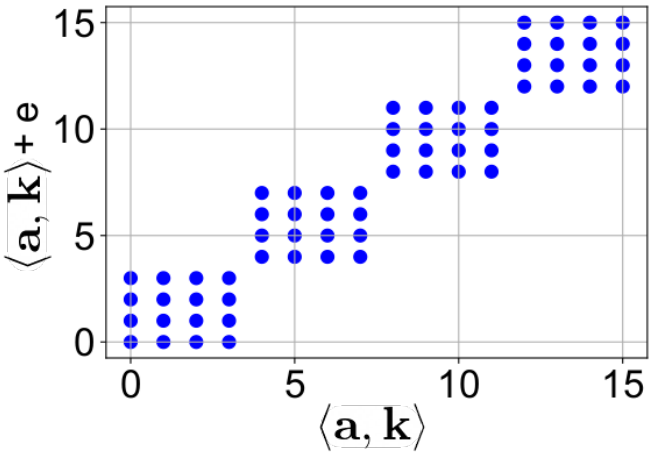
Regular LWE



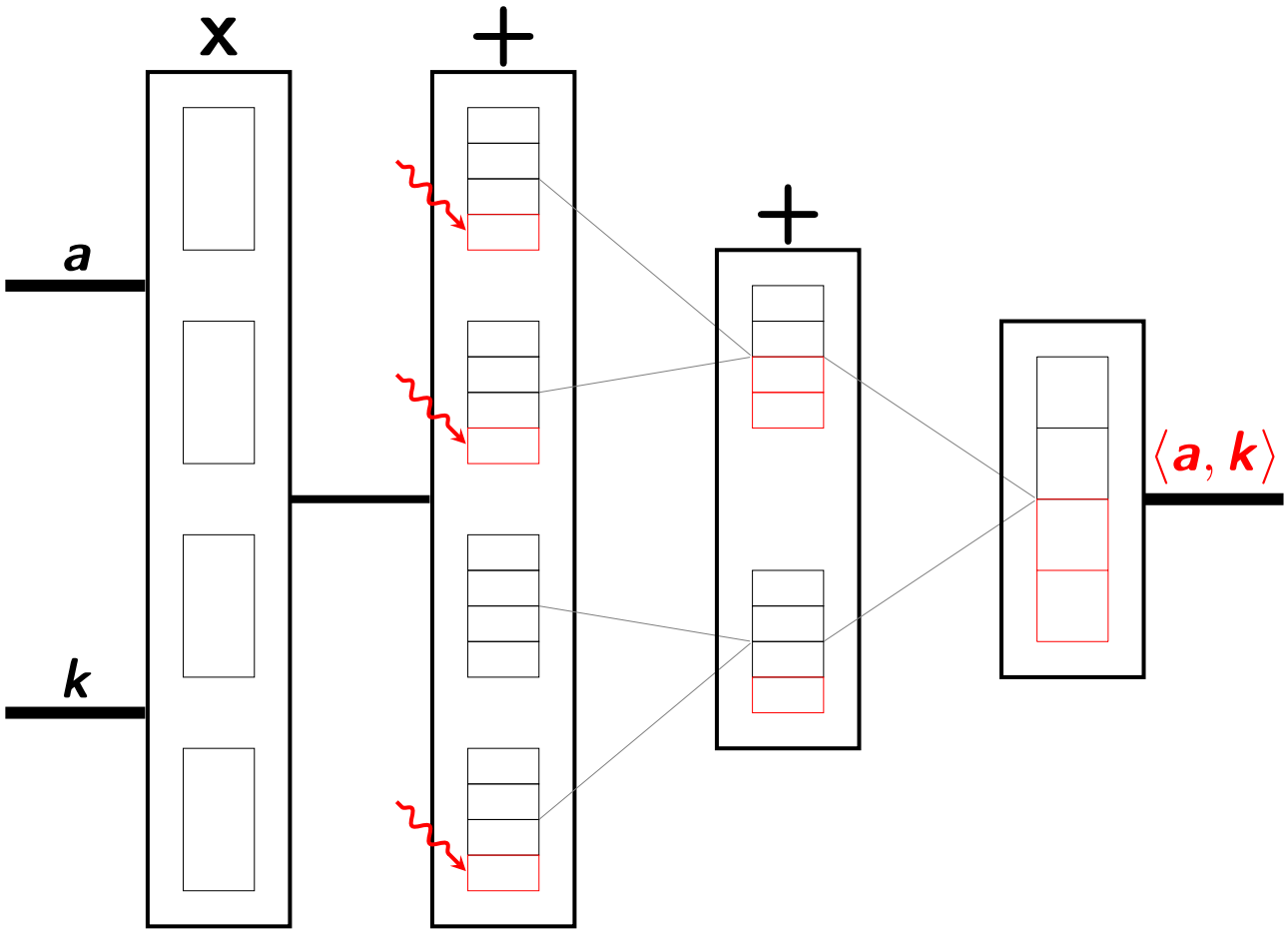


LWPE$_A$

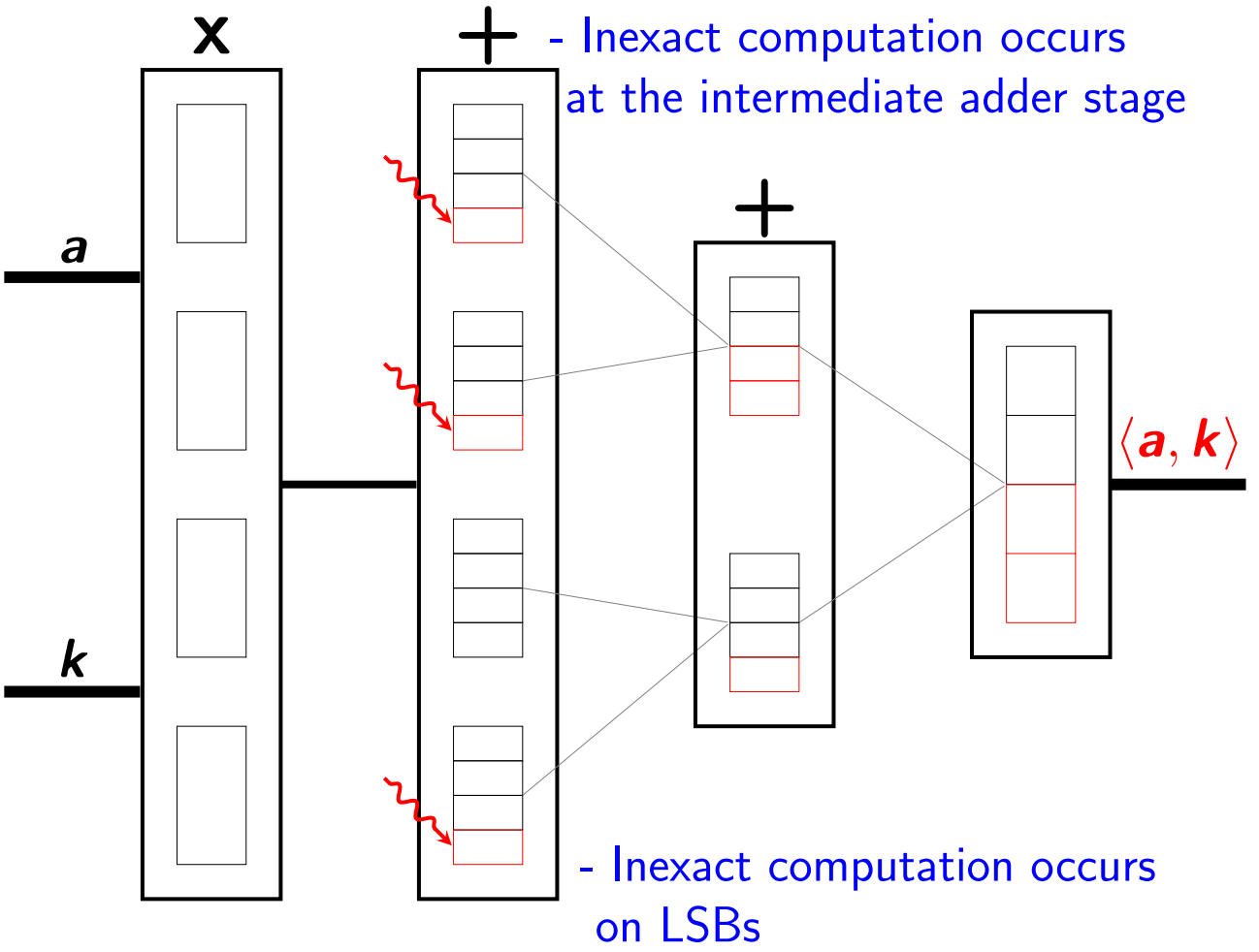




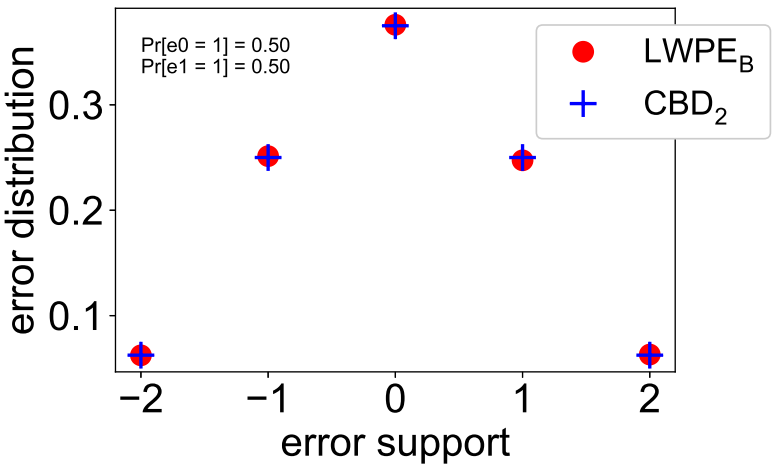
LWPE_B



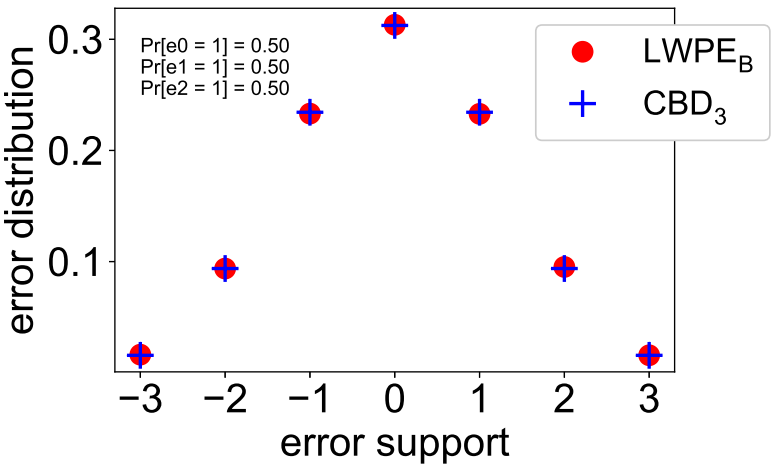
LWPE_B



Results



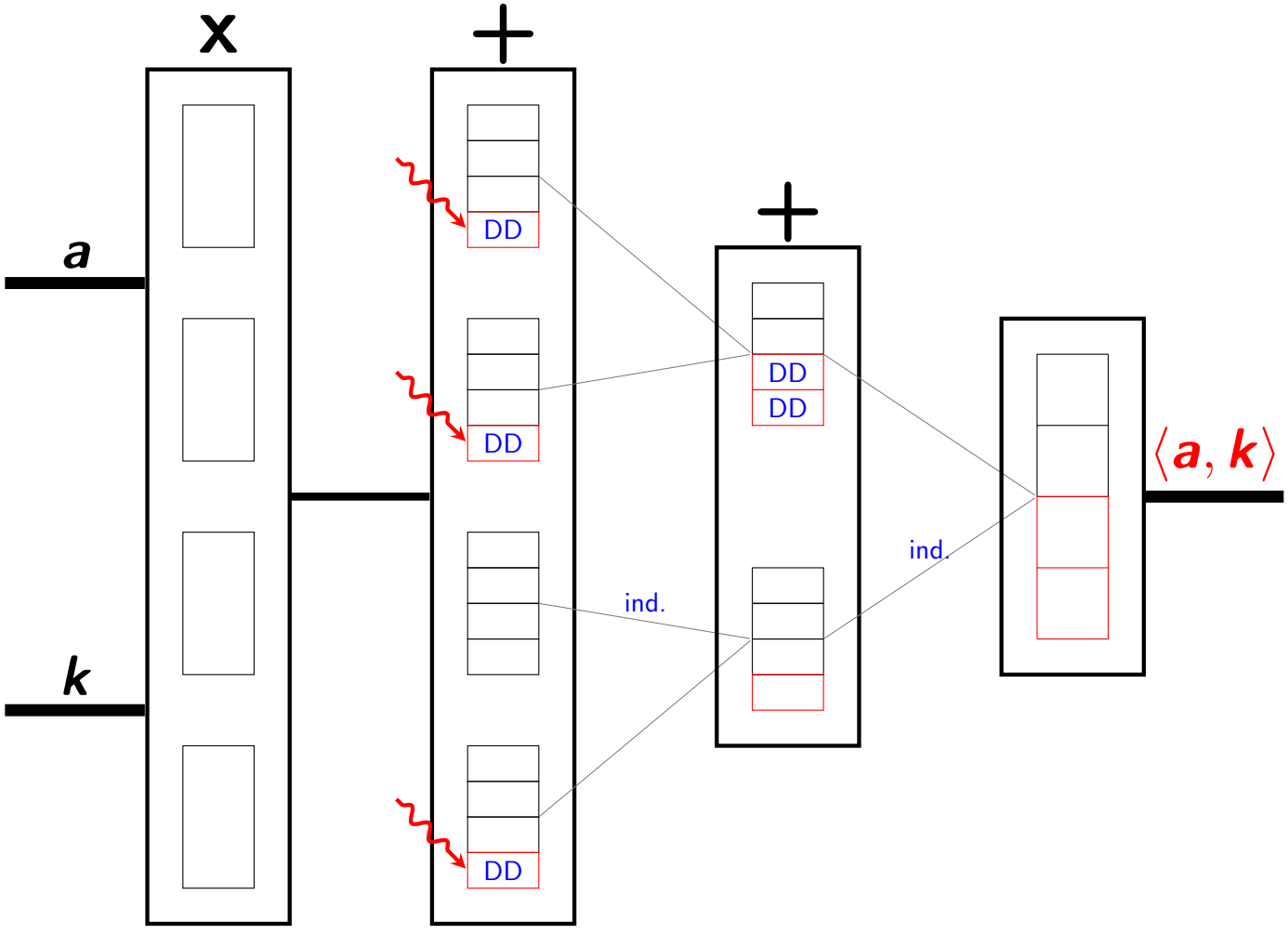
CBD$_2$



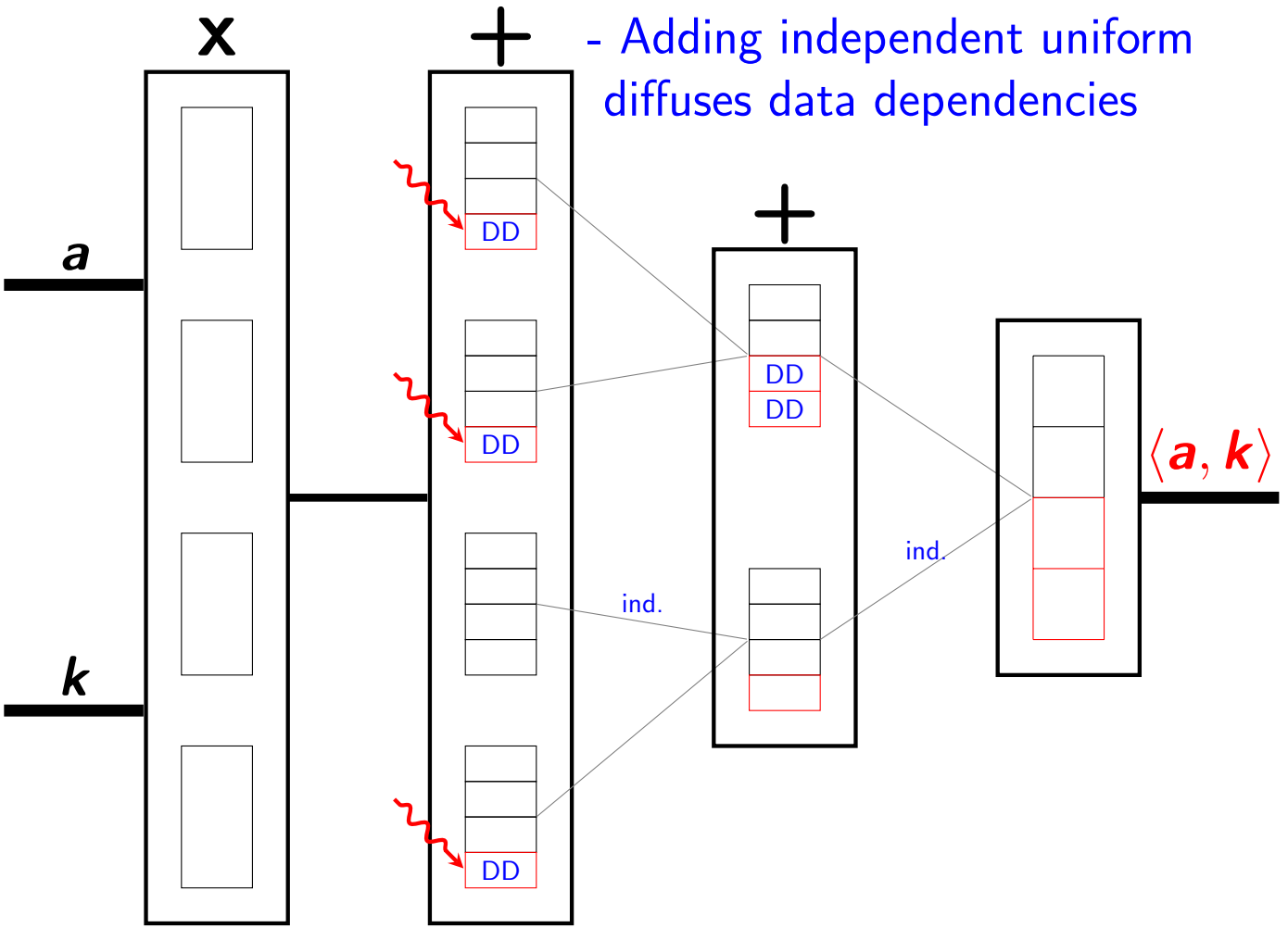
CBD$_3$

LWPE$_B$: Error distribution

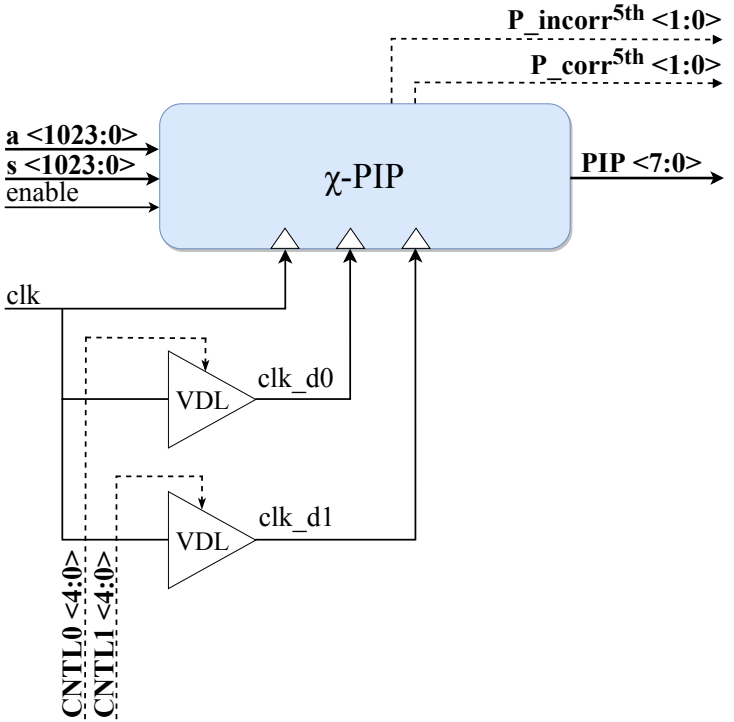
Good news - physical data dependencies



Good news - physical data dependencies



FPGA prototype and data dependencies



LWPE FPGA prototype (dashed lines are only for configuration and testing).

Empirical verification that **data dependencies** cannot be observed

Conclusion - What did we gain?



Interesting potential against leakage
(thanks to the key-homomorphic nature of the operation)

$$\langle \boldsymbol{a}, \boldsymbol{k} \rangle = \langle \boldsymbol{a}, \boldsymbol{k}_0 \rangle + \langle \boldsymbol{a}, \boldsymbol{k}_1 \rangle + \cdots + \langle \boldsymbol{a}, \boldsymbol{k}_{\boldsymbol{d}-1} \rangle$$

Conclusion - Next steps

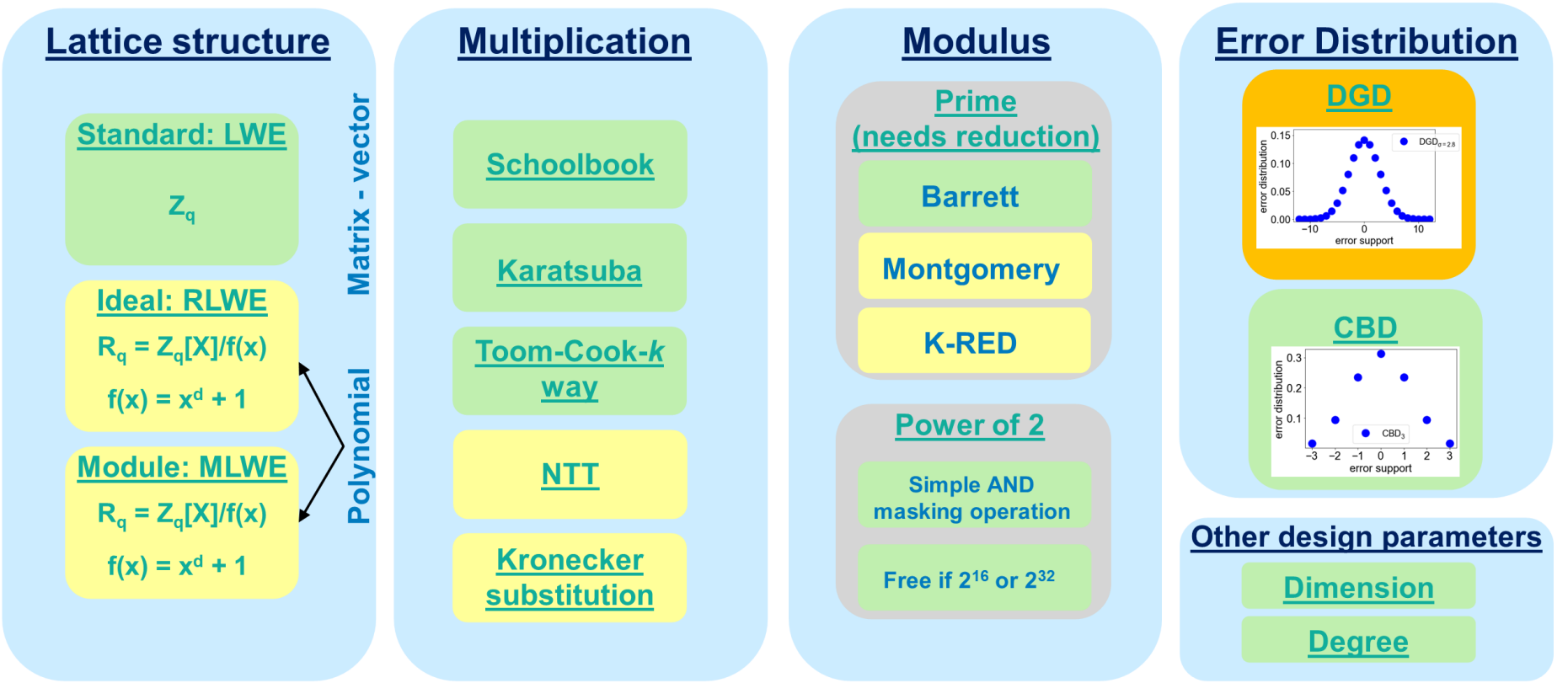


Figure: LWPE design space

green: applicable as is, yellow: small adjustments required

orange: challenging to achieve